RM Procedure 2016_019

May 16th

2016

Machine Remote Diagnosis for RD Ver 1.0

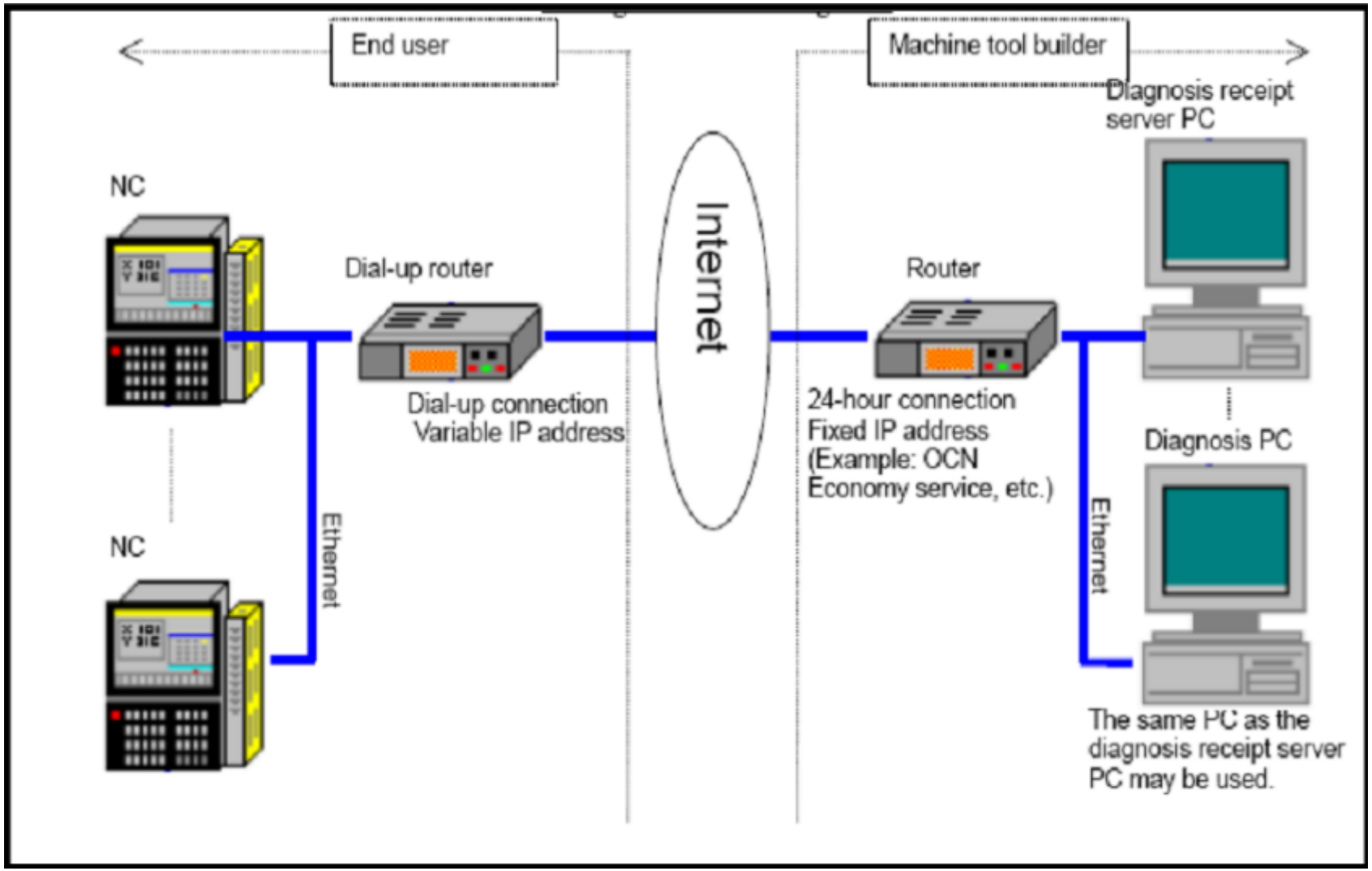
[Applicable for
All Robodrill with FS18iB~31iB]

Machine Remote Diagnosis for Robodrill with FS31*i*-A/B

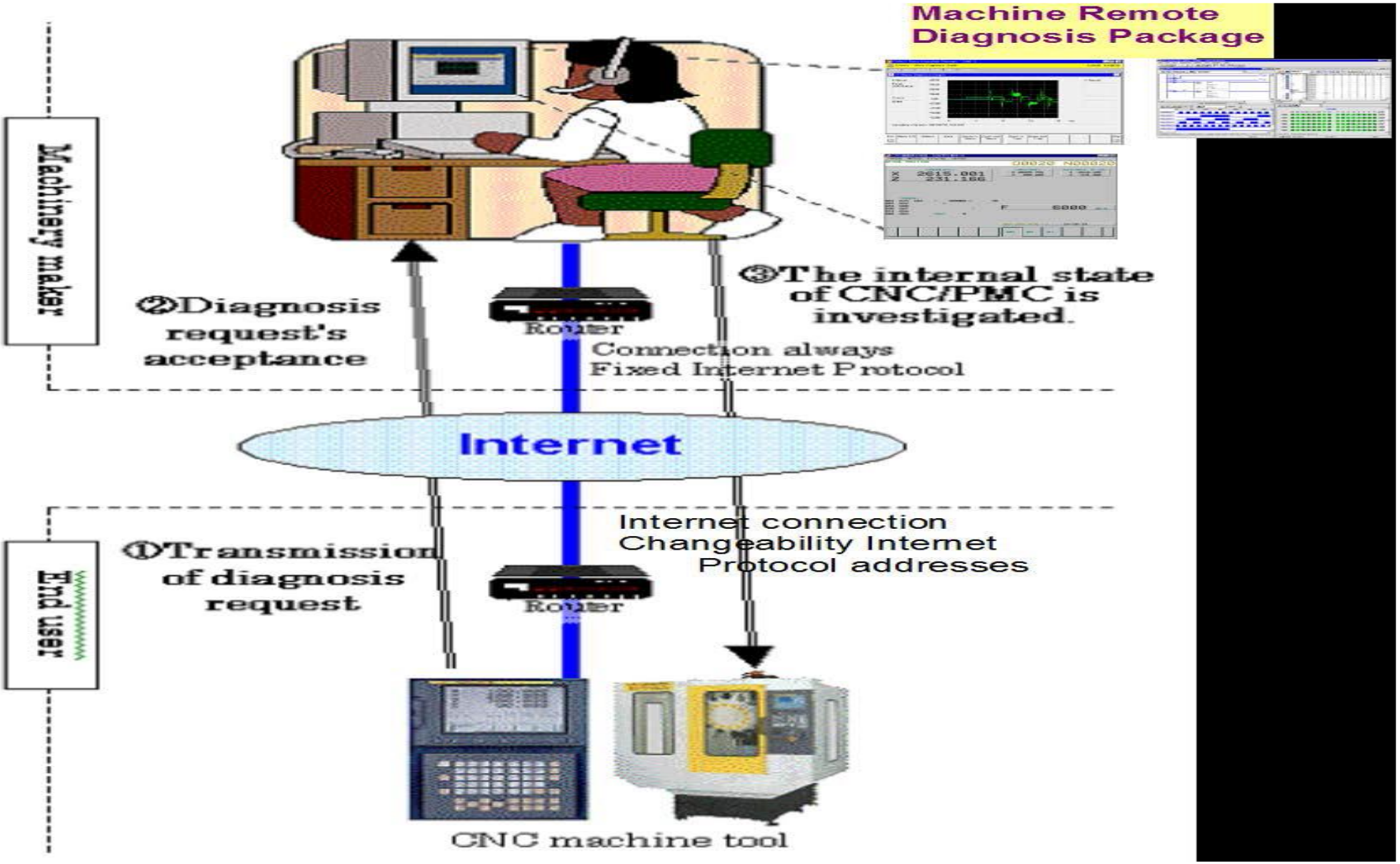
MRD – Introduction

- Machine Remote Diagnosis (MRD) package is a tool designed for a Machine Tool Builder (MTB) to provide support on maintenance of the machines easily from a remote location
- With the MRD functions, checking of the internal CNC status, Ladder program editing and other operations can be performed as necessary by using a personal computer through a LAN from Remote location
- By using a internet, the state of a CNC machine tool installed far from the service centre can be monitored and its data can be modified
- Form End-User information such as alarm/message data, parameters and programs can be transferred to the service centre and viceversa . So that speedy/timely maintenance and service are possible
- MRD function in the NC works on Embedded Ethernet or Fast Ethernet hardware with necessary NC option and PC softwares.

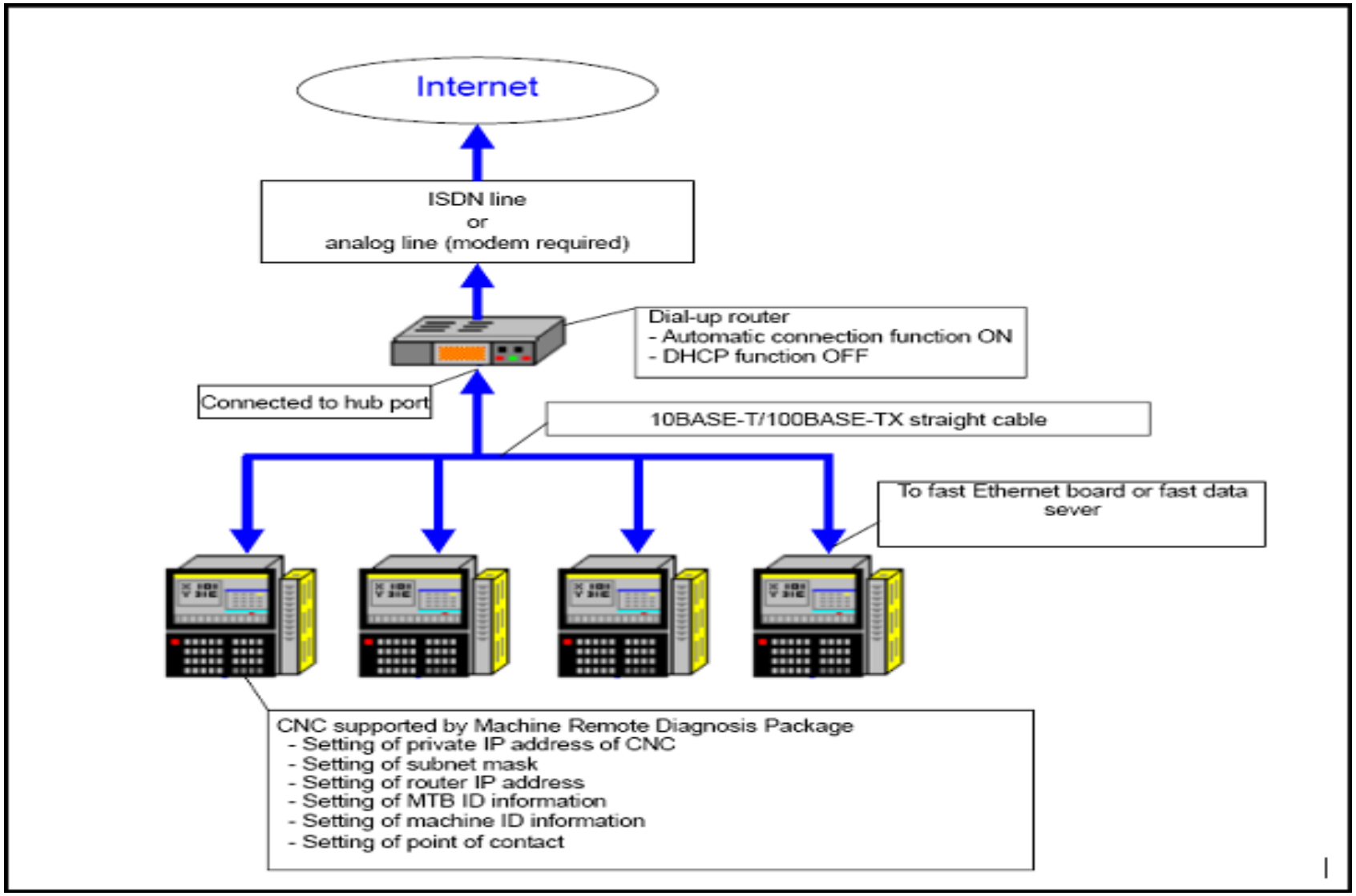
MRD – Basic Structure



MRD – Basic Structure



MRD – End User Side Setup



MRD – Basic Requirements

On CNC Side:

1. Controller with Embedded or Fast Ethernet port is a must
2. Necessary NC, Ethernet software supporting MRD (Please refer necessary controller description manual for information)

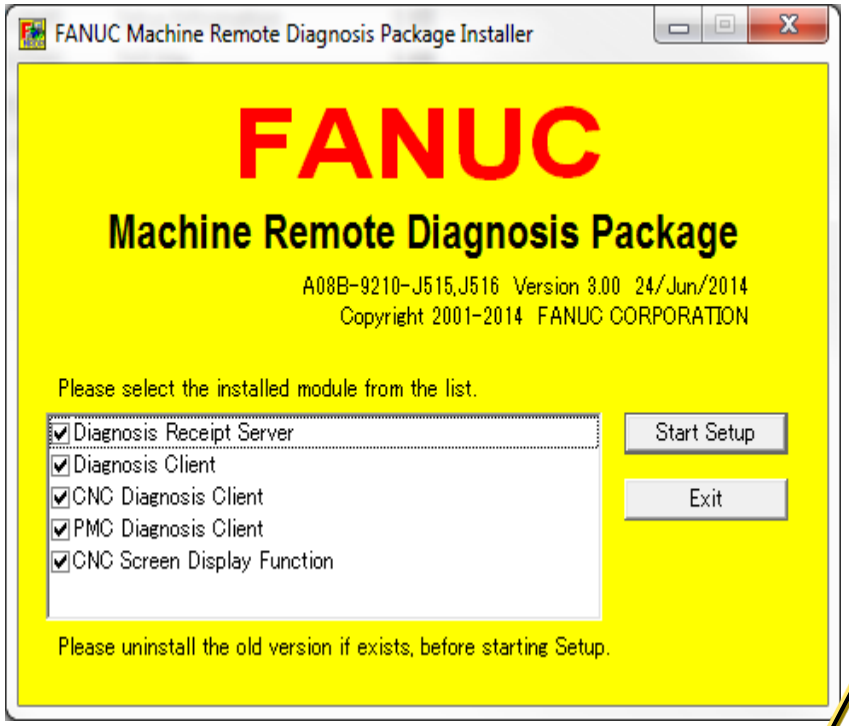
   Likely:

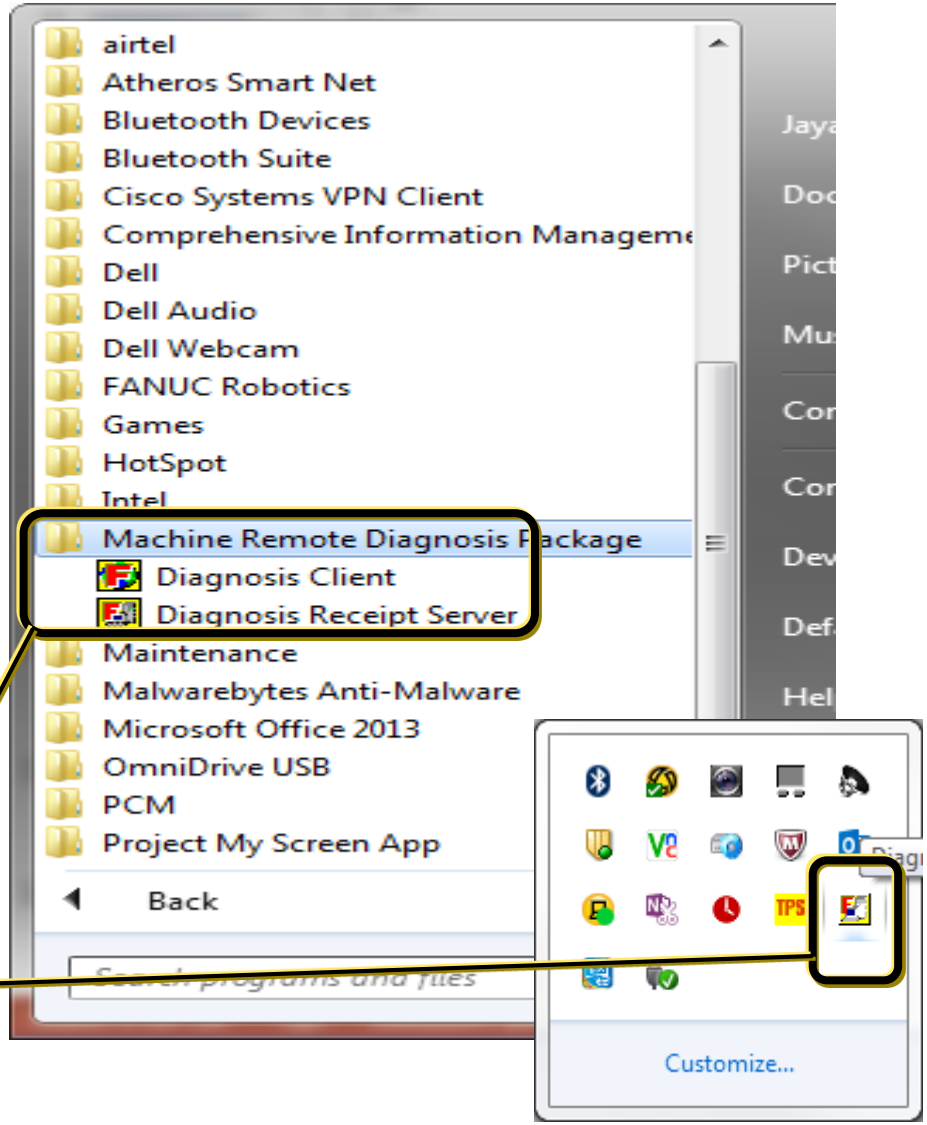
   – A02B-0XXX-S707 → Ethernet Function
   – A02B-0XXX-S884 → Dual Screens of CNC Screen display Function
   – 658E → NC Software for Ethernet Function

On Factory Side:

1. PC supporting Ethernet feature - Cross for single NC and straight for connection through HUB or Router
2. MRD software (A08B-9210-J515) loaded to the PC
3. Ethernet Switch or Hub to connect the multiple machine in the shop floor to MRD PC Server
4. Router with dedicated Internet connection incase of accessing machines from outside factory
5. MRD Package Operator Manual → B-63734EN

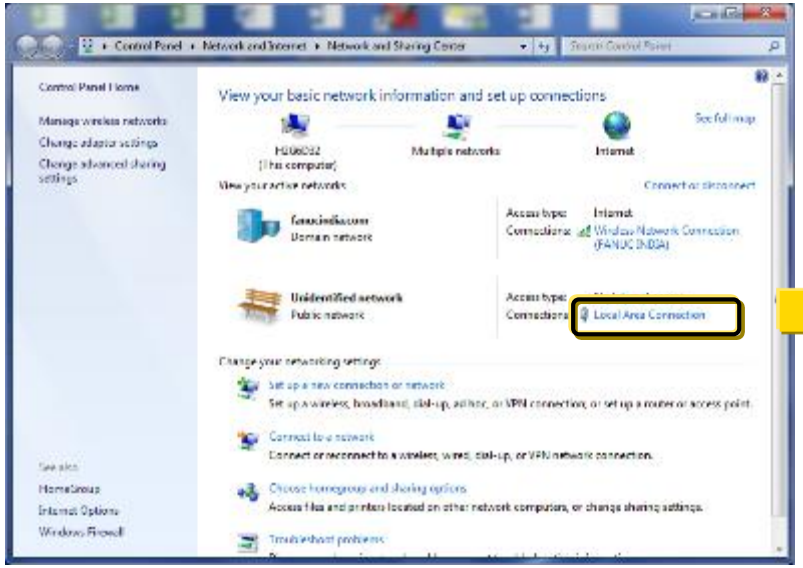
MRD – PC Software Installation

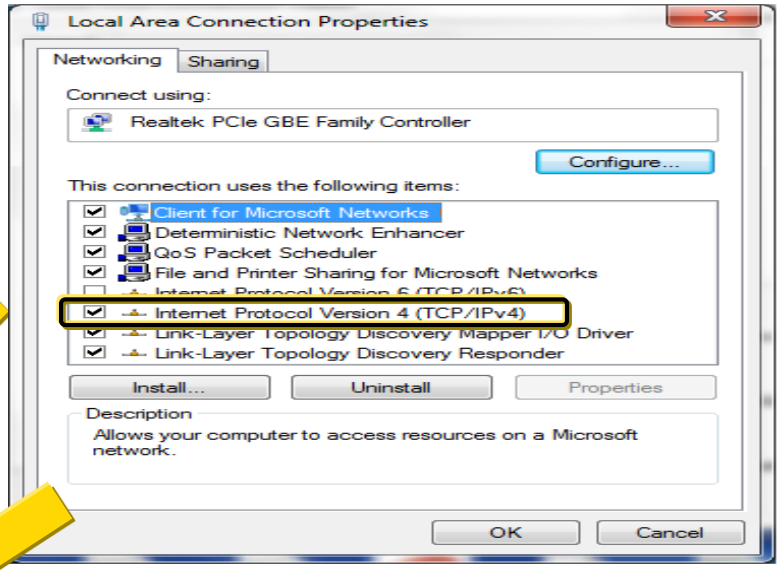


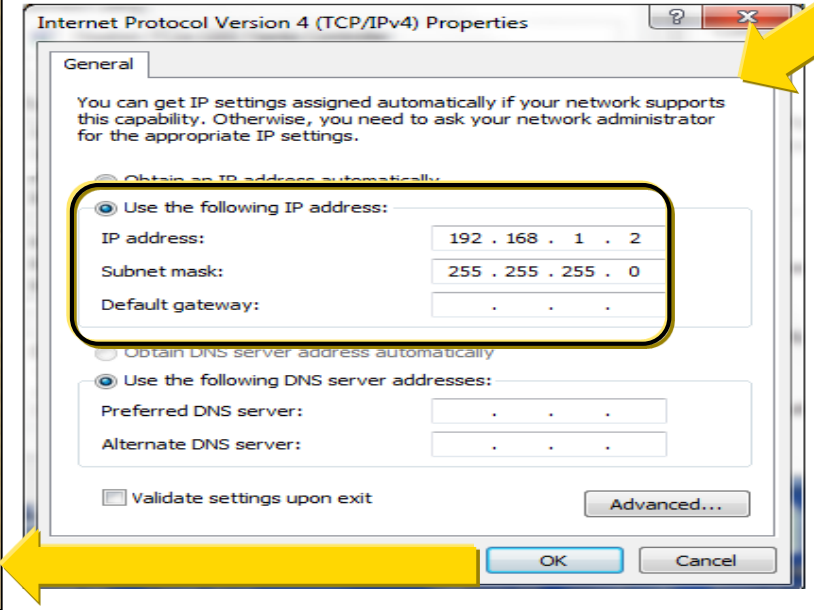


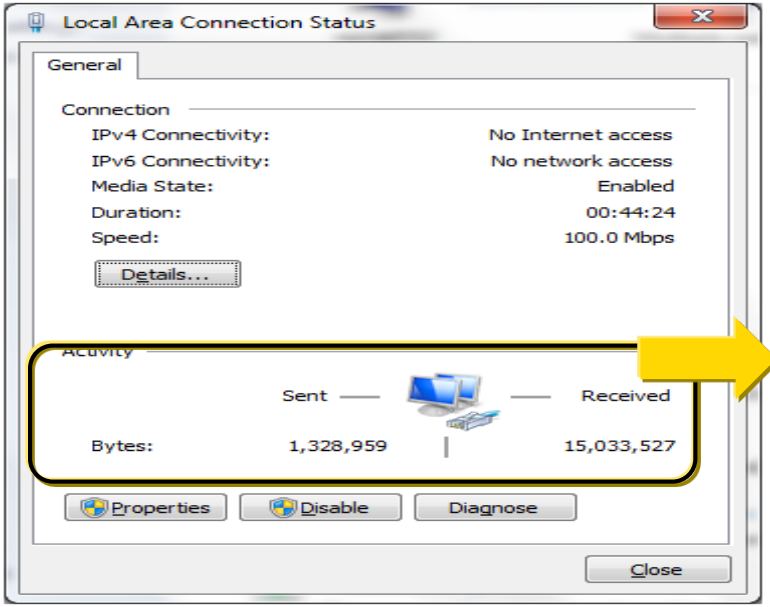
Please setup the PC with MRD software:
A08B-9210-J515 – New installation
A08B-9210-K515#Z – Upgrading S/w for old Version
After installation please check for **Start Menu** and **Icon Tray** for software packages notification as encircled

MRD – PC Side IP Setting





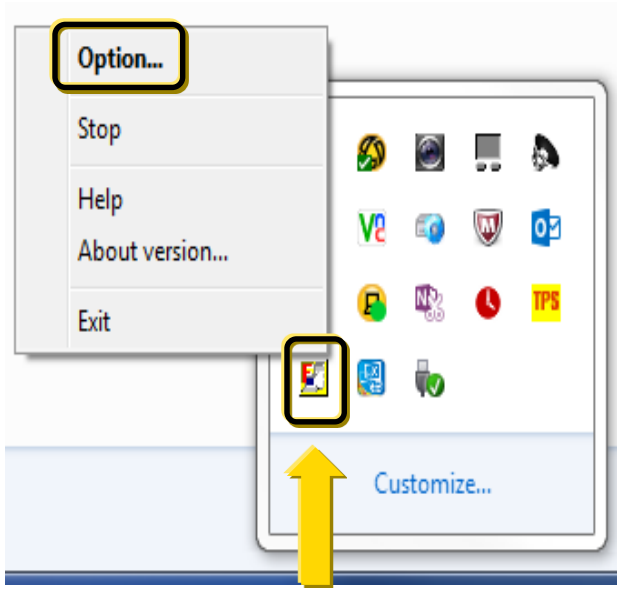




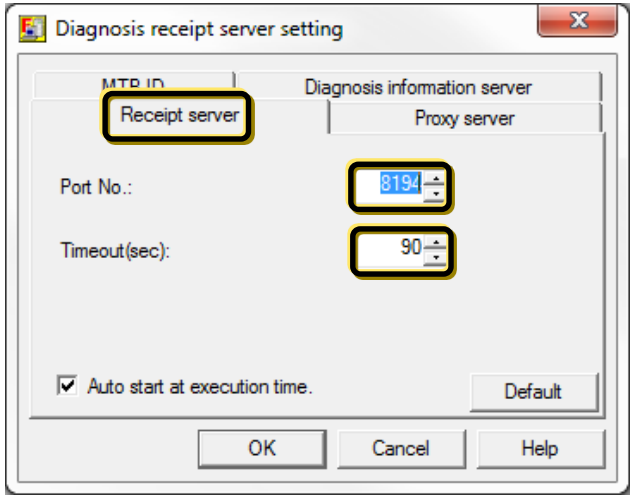
After Setting all the above circled item than press **OK** to record the data entered

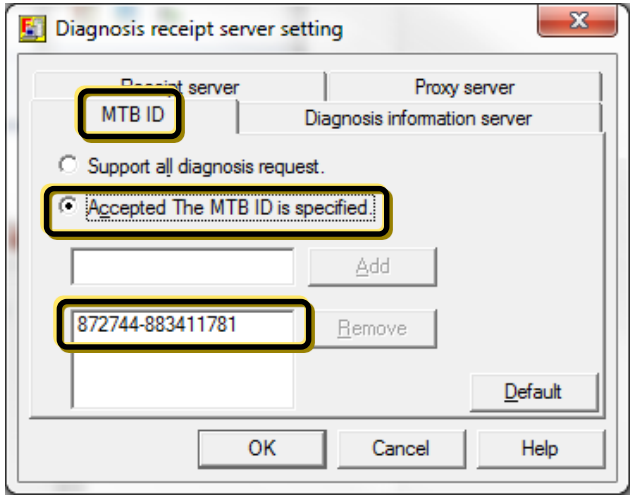
The counter status denotes the communication is active

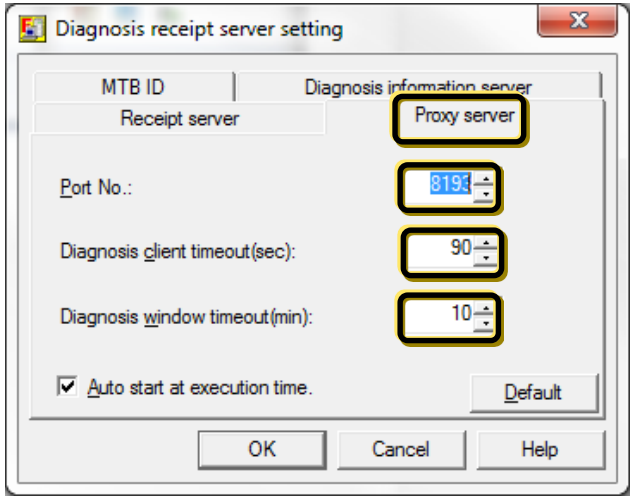
MRD – PC Side PRM Setting

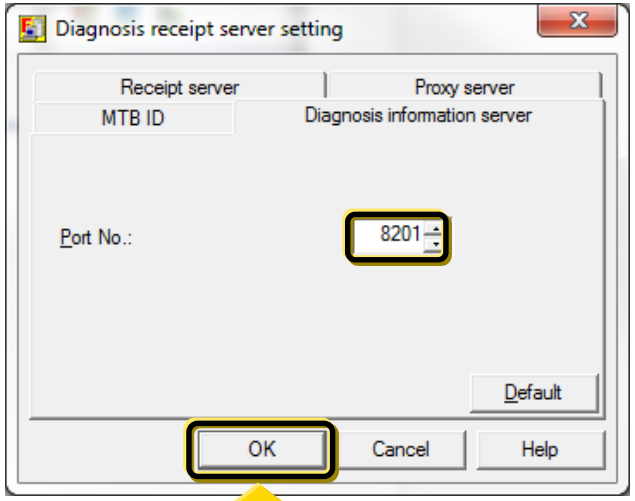


Click on the MRD Icon in tray to access the MRD setting as shown







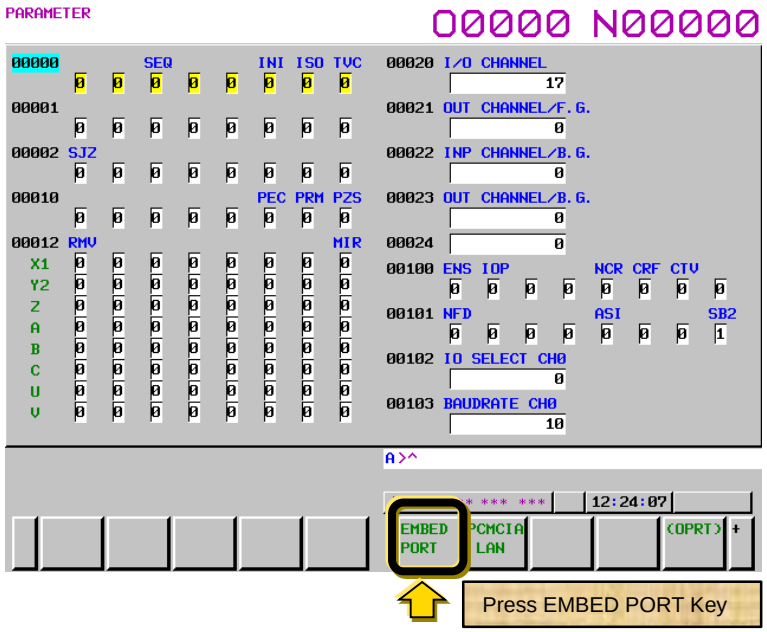


After Setting all the above circled item than press **OK** to record the data entered

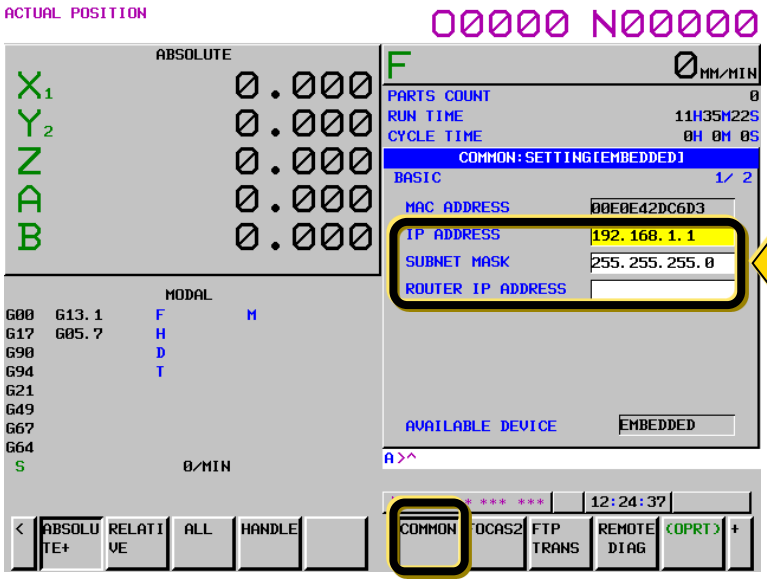
MRD – CNC PRM Setting

1. Please load the Ethernet NC Software to controller through Boot Menu according to controller specification likely as below:
   - 658E or later Version → NC Software for Ethernet Function
2. Set the NC Option parameters through OPTION INPUT menu or ALL I/O
   - A02B-0XXX-S707 → Ethernet Function
   - A02B-0XXX-S884 → Dual Screens of CNC Screen display Function
3. On NC side please change the below parameters (for description please refer CNC Parameter manual:
   - PRM 24 = 10
   - PRM 905#1 = 0
   - PRM 3206#6 = 1
   - PRM 8706#6 = 0
   - PRM 14880#1&5 = 0
   - PRM 14882#0 = 1 (In-case Embedded Ethernet is used for MRD)
4. Reference manuals:
   - B-64605EN – Embedded Ethernet function
   - B-64014EN – Setting the MRD Function
   - B-63734EN – MRD Package Operator's Manual
   - B-63644EN – Fast Ethernet Board Operator Manual

MRD – CNC Side IP Setting (1/2)

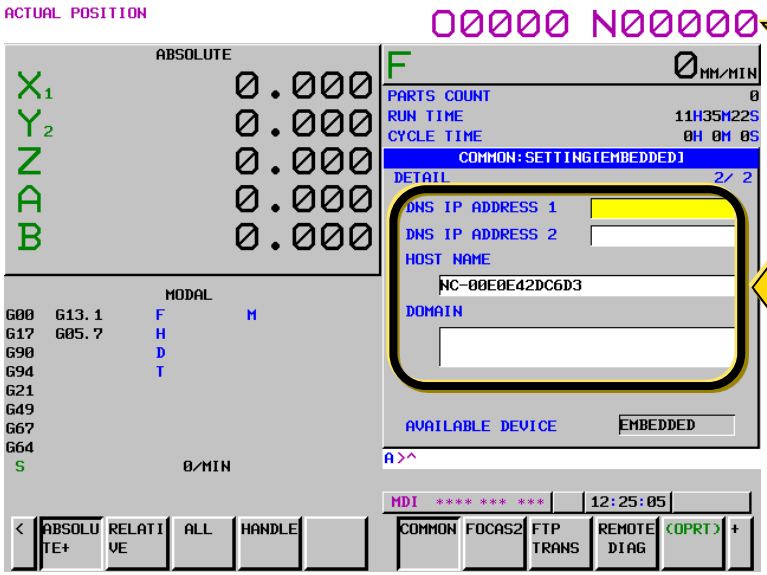


Press EMBED PORT Key



Press enter IP address as per field requirement

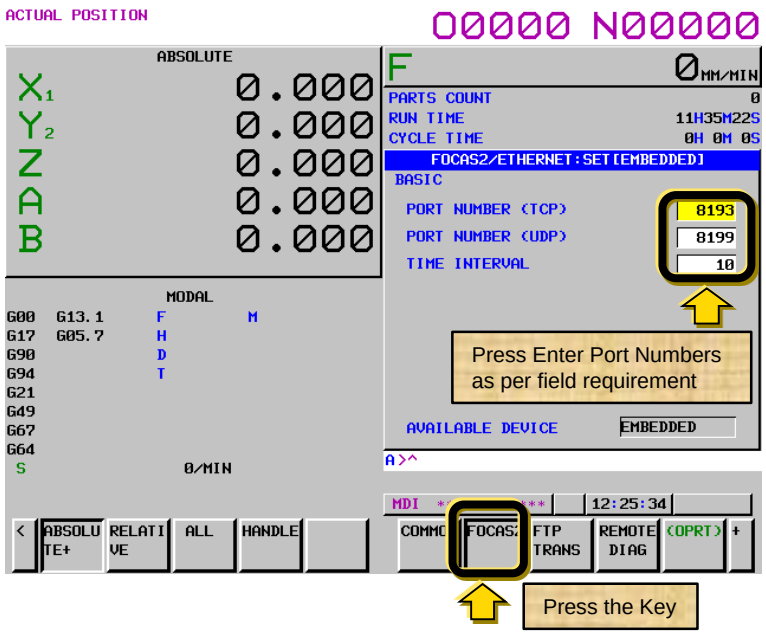
Press **Page Down** Key



Input Host Name as per the field required. Not mandatory for Non-DNS operation

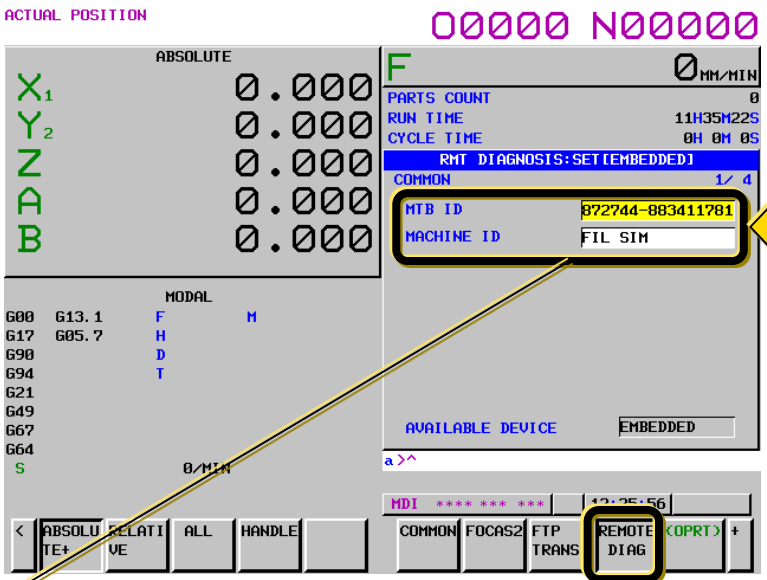
- Press **SYSTEM** Key and Right Arrow Softkey several times to get **EMBED PORT**
- IP address should be inline at both PC & CNC side
- Router IP must be HUB IP address
- Rest IP are as per the field requirements
- To delate the IP address please use combination key **SHIFT + SP + INPUT**

MRD – CNC Side IP Setting (2/2)



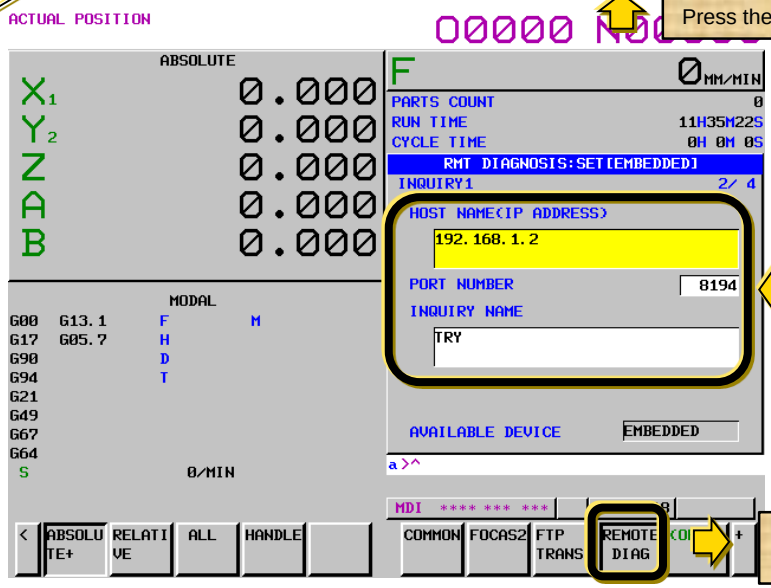
Press Enter Port Numbers as per field requirement

Press the Key



Press Enter MTB ID as per MRD PC Setting and Machine ID as per the field required

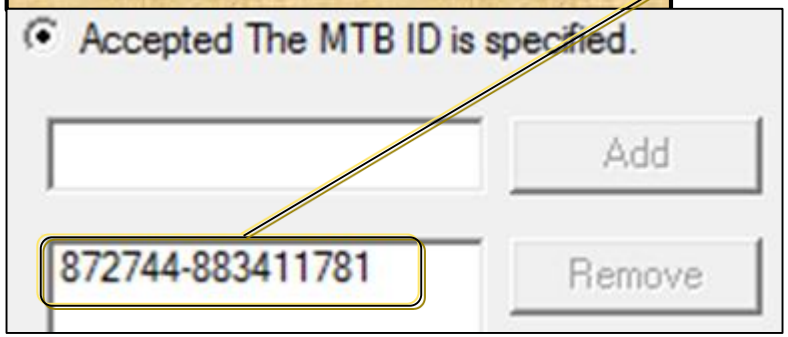
Press the Key



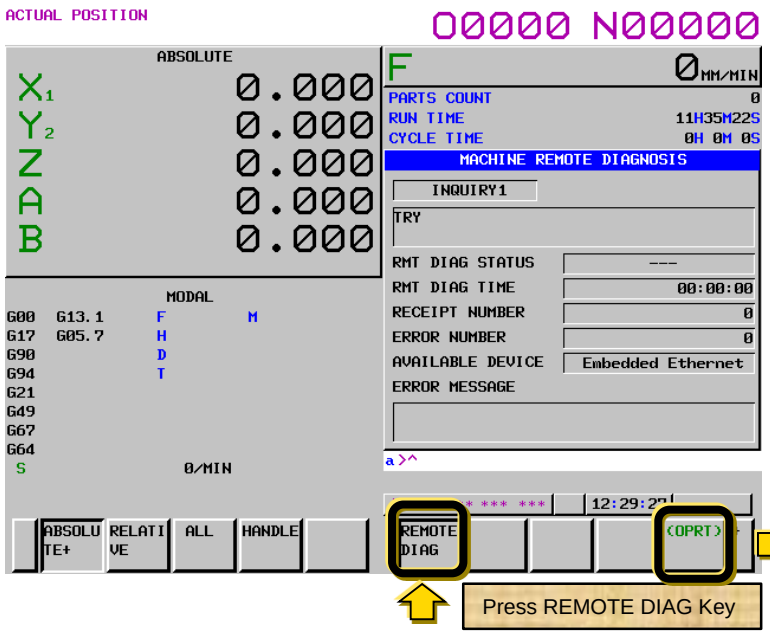
Press Enter **Host IP** as per the connected PC. **Port Number** and **Inquiry Name** as per the field requirement

Press the Key to get screen

- MTB ID must be same the MRD option setting on PC Side

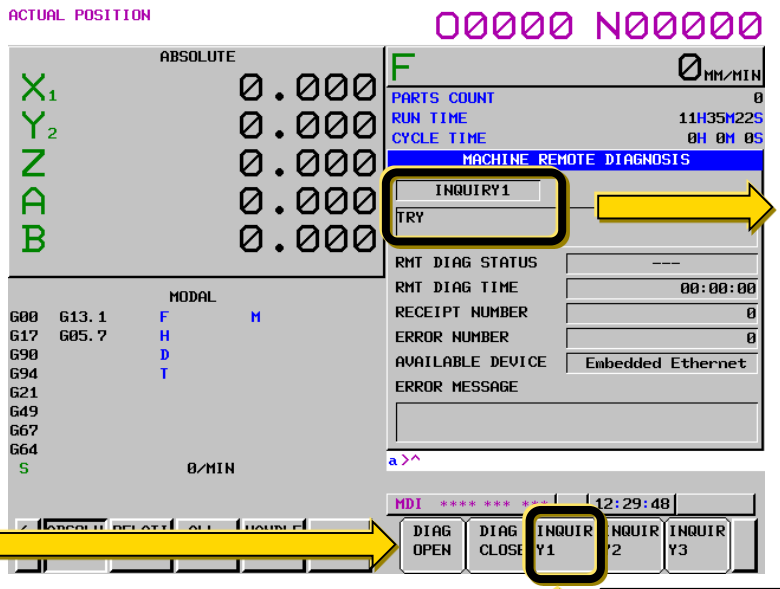


MRD – CNC Side Start



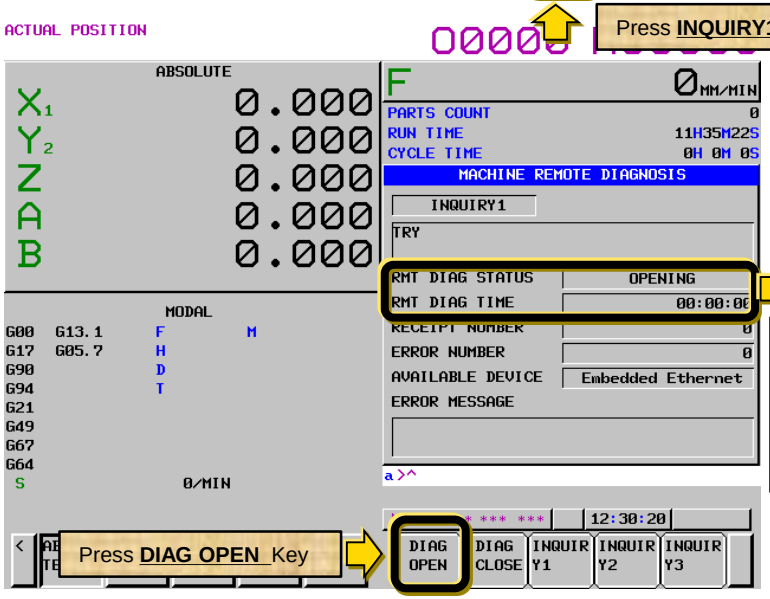
Press REMOTE DIAG Key

Press **OPRT** Key to SUB-MENU



Indicates INQUIRY1 is active

Press **INQUIRY1** Key

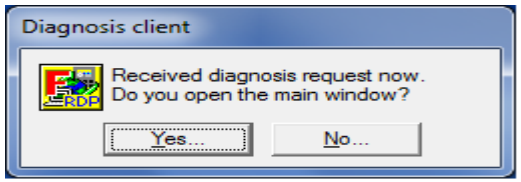


Press **DIAG OPEN** Key

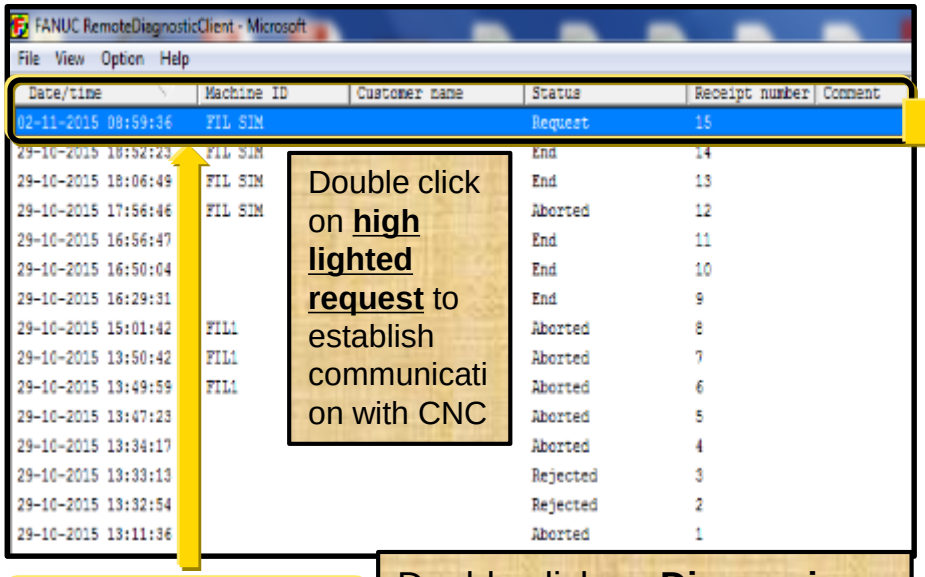
OPENING Indicates INQUIRY1 request send to PC

- Press **SYSTEM** Key and Right Arrow Softkey several times to get **REMOTE DIAG**
- Set **INQUIRY1** by pressing Softkey
- Than Press **DIAG OPEN** to establish communication with PC

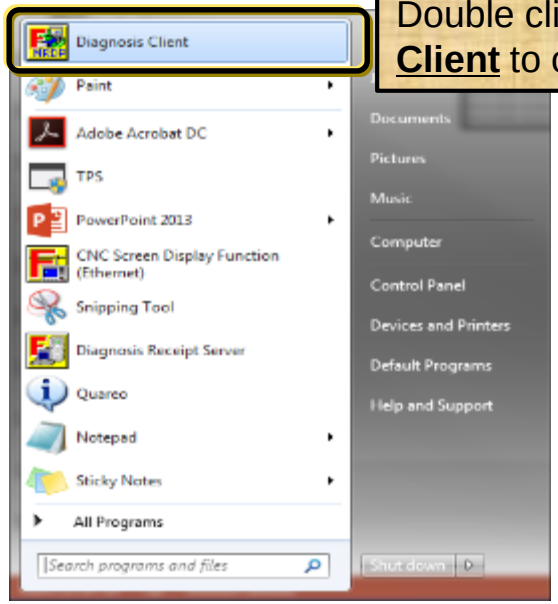
Below screen appears on PC side, if Diagnostic Client is already open/running



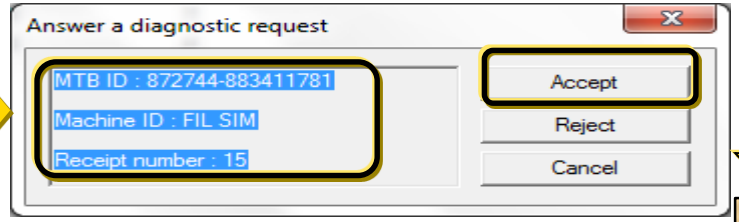
MRD – PC Side Start



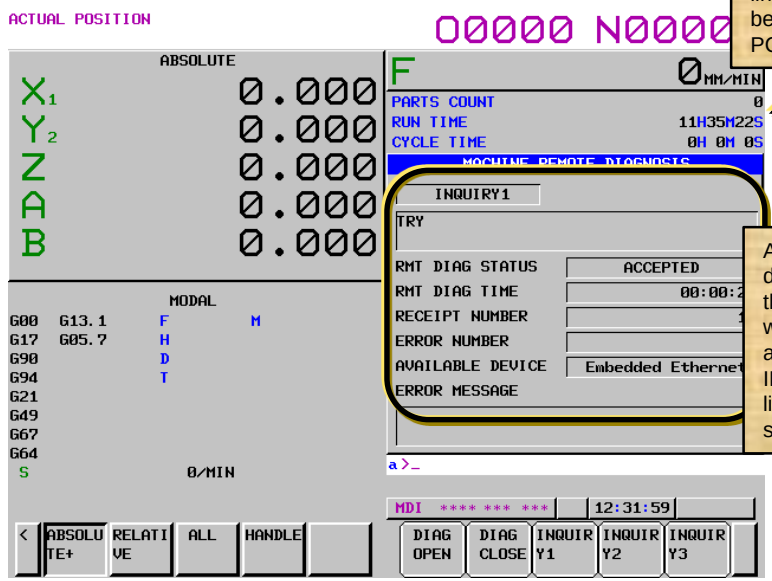
Double click on **high lighted request** to establish communication with CNC



Double click on **Diagnosis Client** to open above MENU



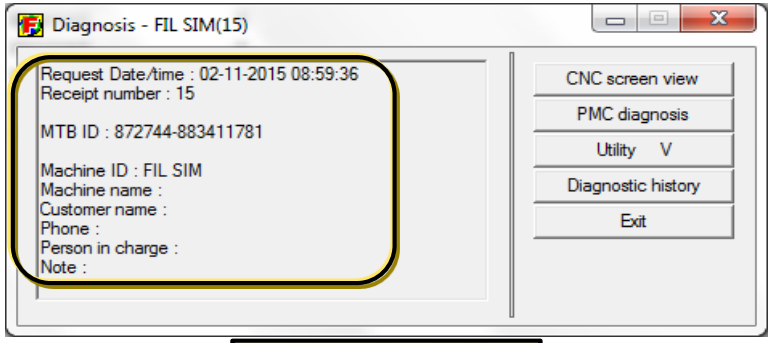
Press ACCEPT to get start link between PC & CNC



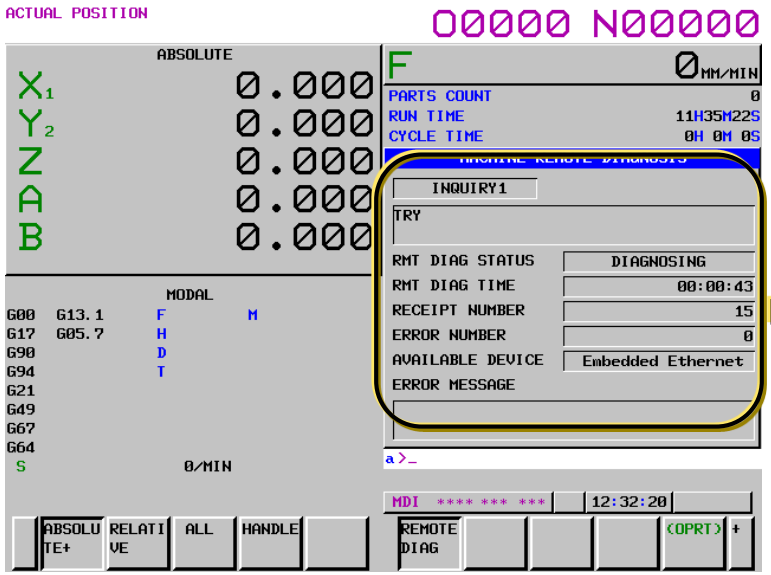
ACCEPTED displayed n the CNC with time and Receipt ID once the link is successful

Maximum of 10 Machines can be diagnosed with one PC software at same time.

MRD – CNC→PC Communication

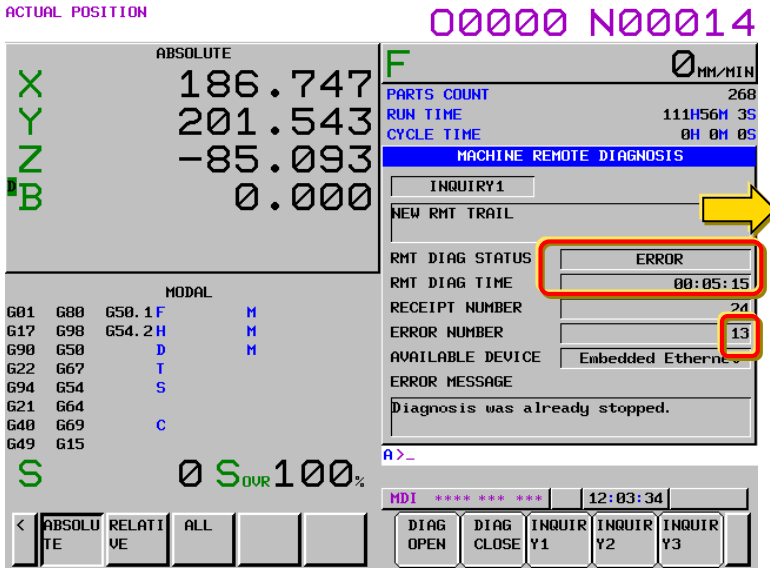


PC Side Display



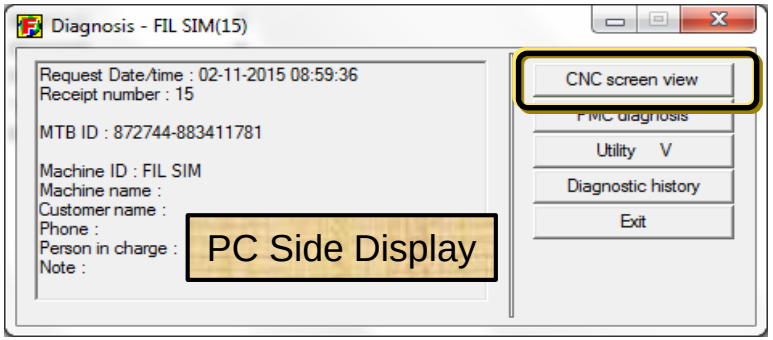
CNC Side Display

- This screens appear on PC and CNC once the link is established
- On PC shows the machine connected to as shown in the above clip
- On CNC shows DIAGNOSISING with RECEIPT NUMBER

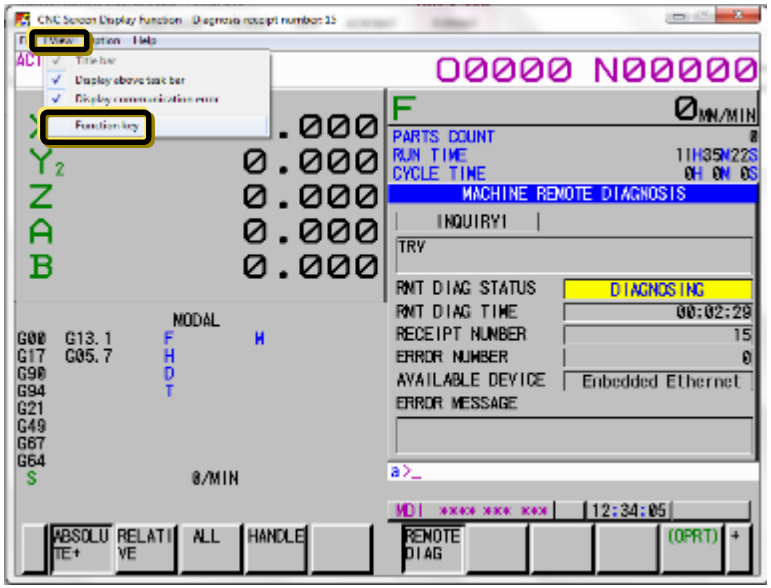


This screens appear on CNC if the link is not established with PC and accordingly Error number is also displayed on the CNC screen

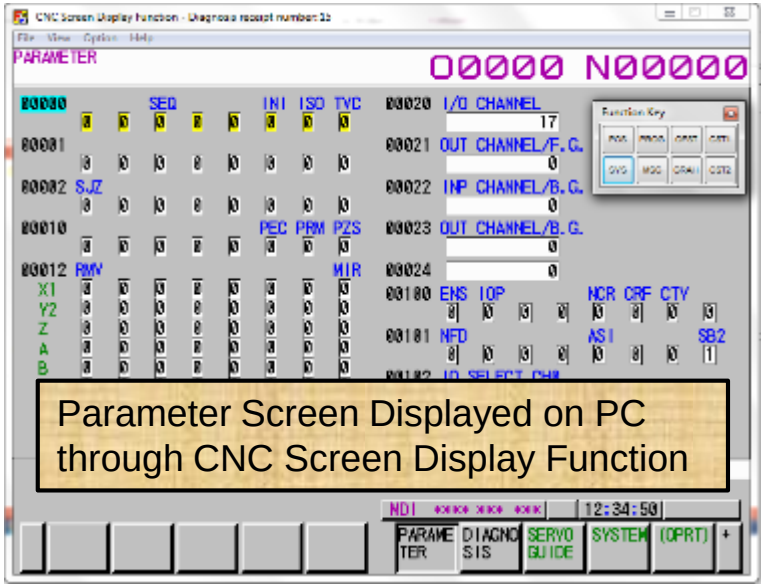
MRD – CNC Screen Viewer (1/3)



Click on **CNC Screen Viewer** to display the CNC screen

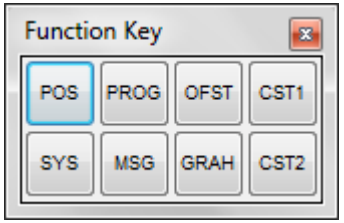


❖ **At most care to be taken during any data change from remote location**



To get Function Key Menu on PC screen:

1. Click **View** on the toolbar
2. Click on **Function Key**

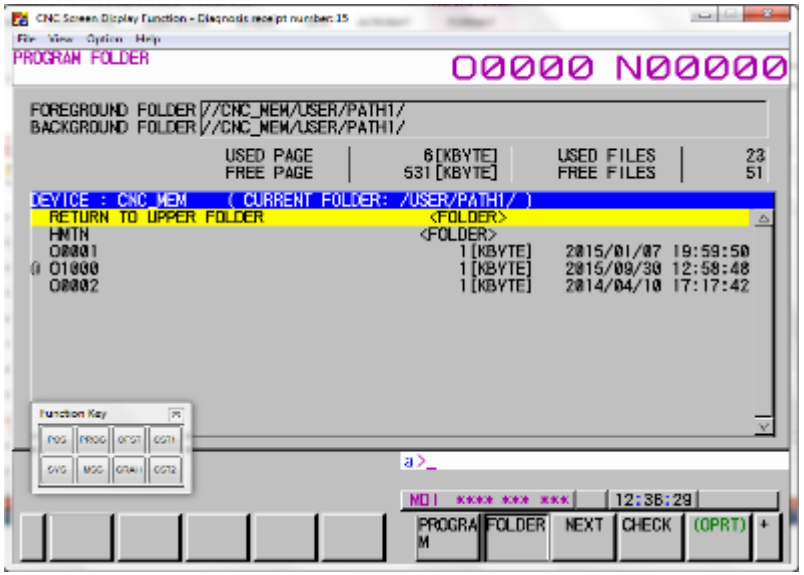


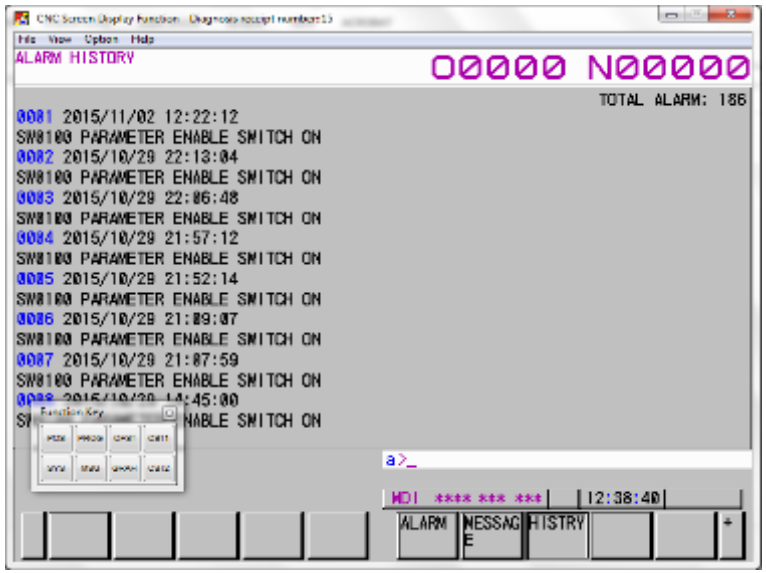
MRD – CNC Screen Viewer (2/3)

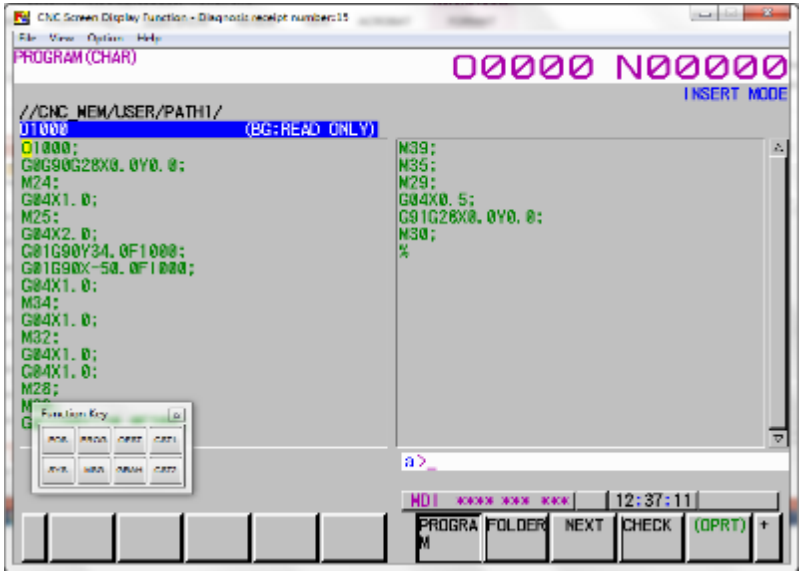
Through CNC Screen display any CNC screen can be viewed and modified the content at PC as shown below:

1. Part Program Screen
2. Alarm Message Screen

❖ **At most care to be taken during any data change from remote location**



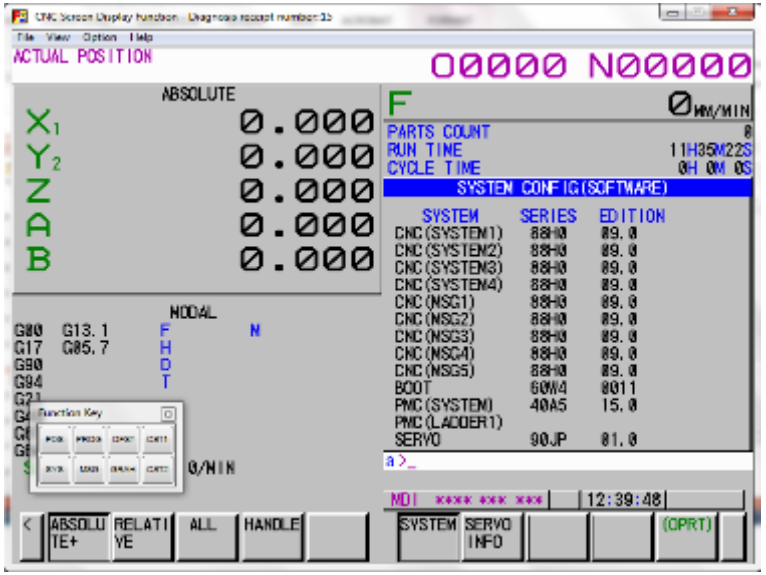


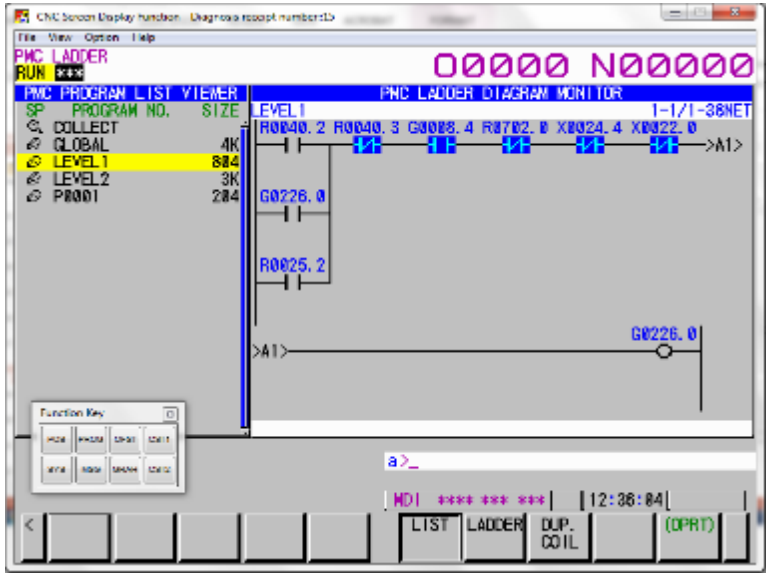


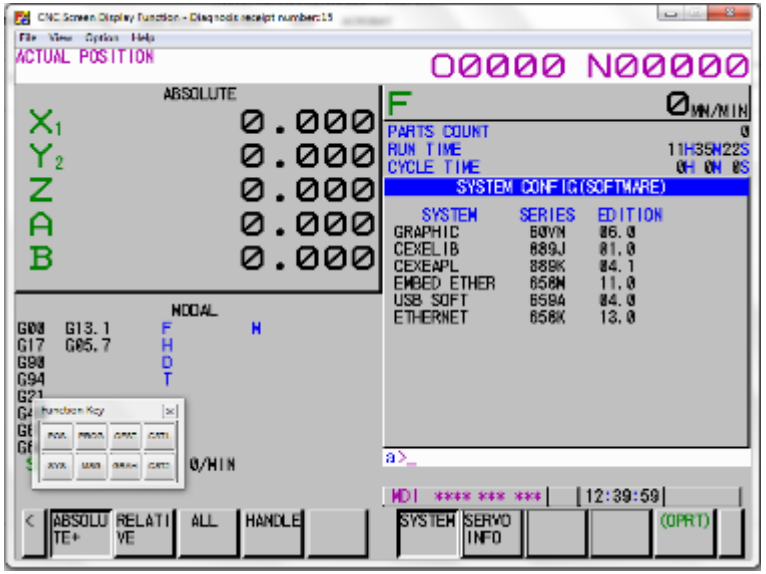
MRD – CNC Screen Viewer (3/3)

Through CNC Screen display any screen related to CNC can be viewed and content can be modified

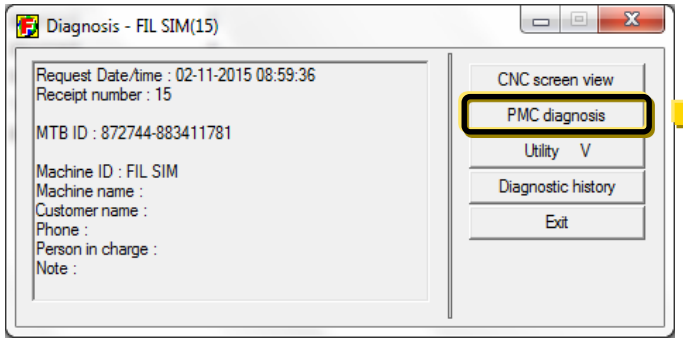
- **At most care to be taken during any data change from remote location**

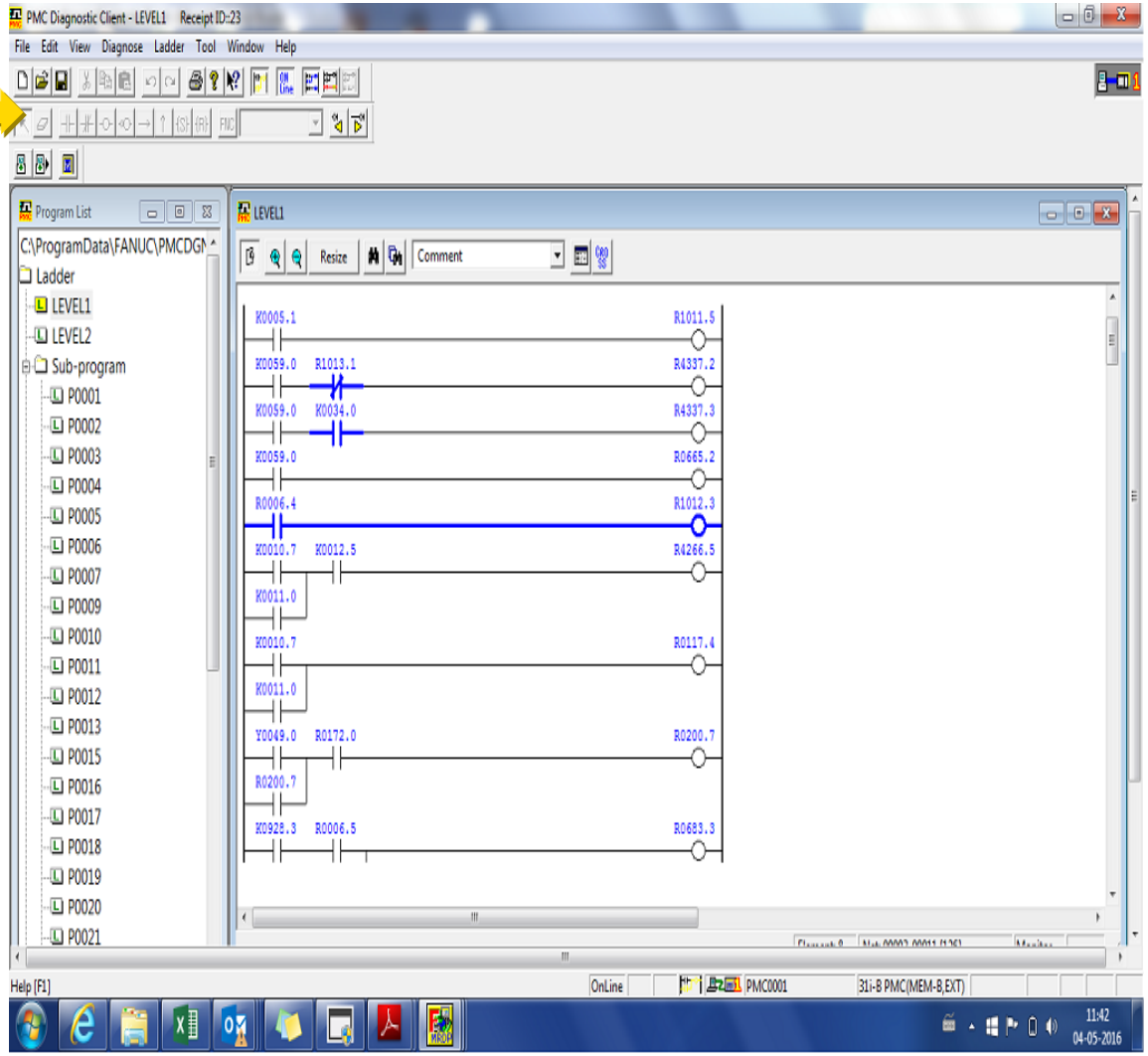






MRD – PMC Diagnosis

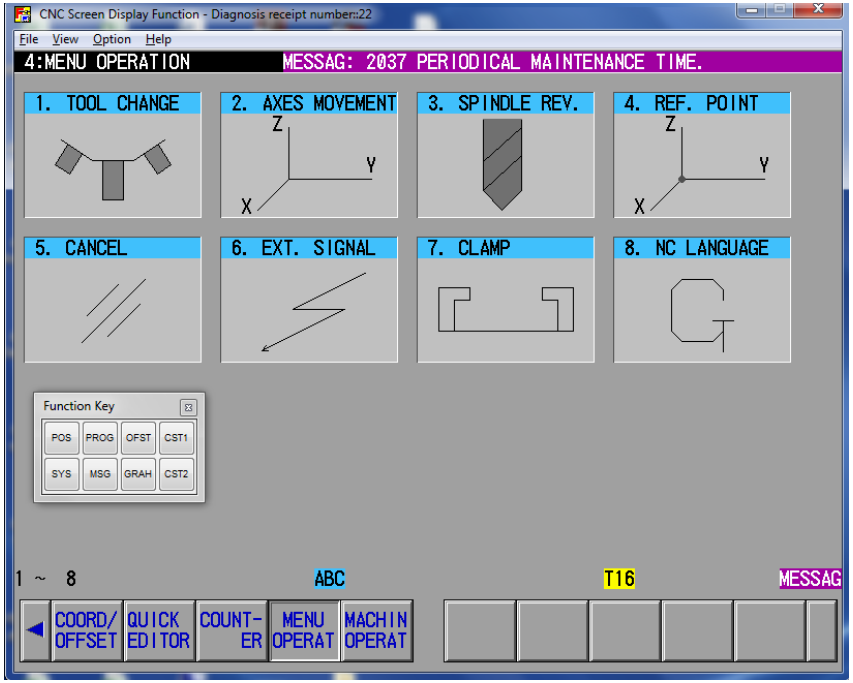


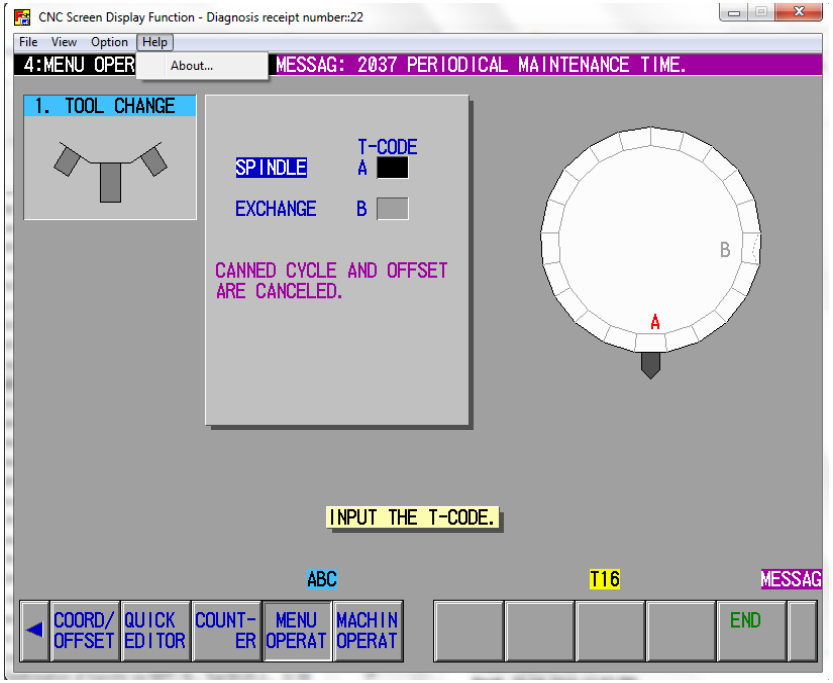


Through PMC Diagnosis Menu: Clicking on **PMC Diagnosis** with display PMC ladder on PC Screen PMC ladder viewing and editing is possible

- **At most care to be taken during PMC data change from remote location**

Quick NC Screen MRD

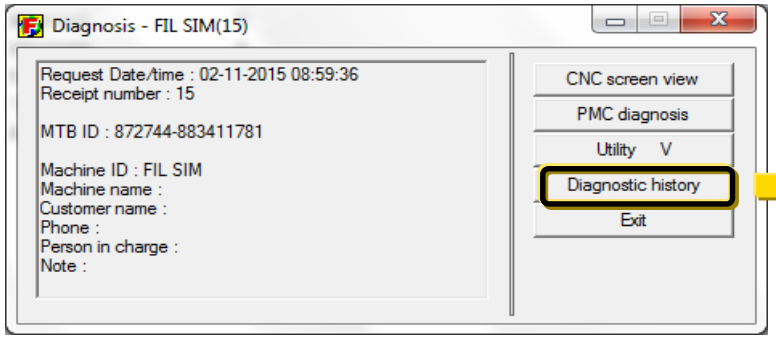


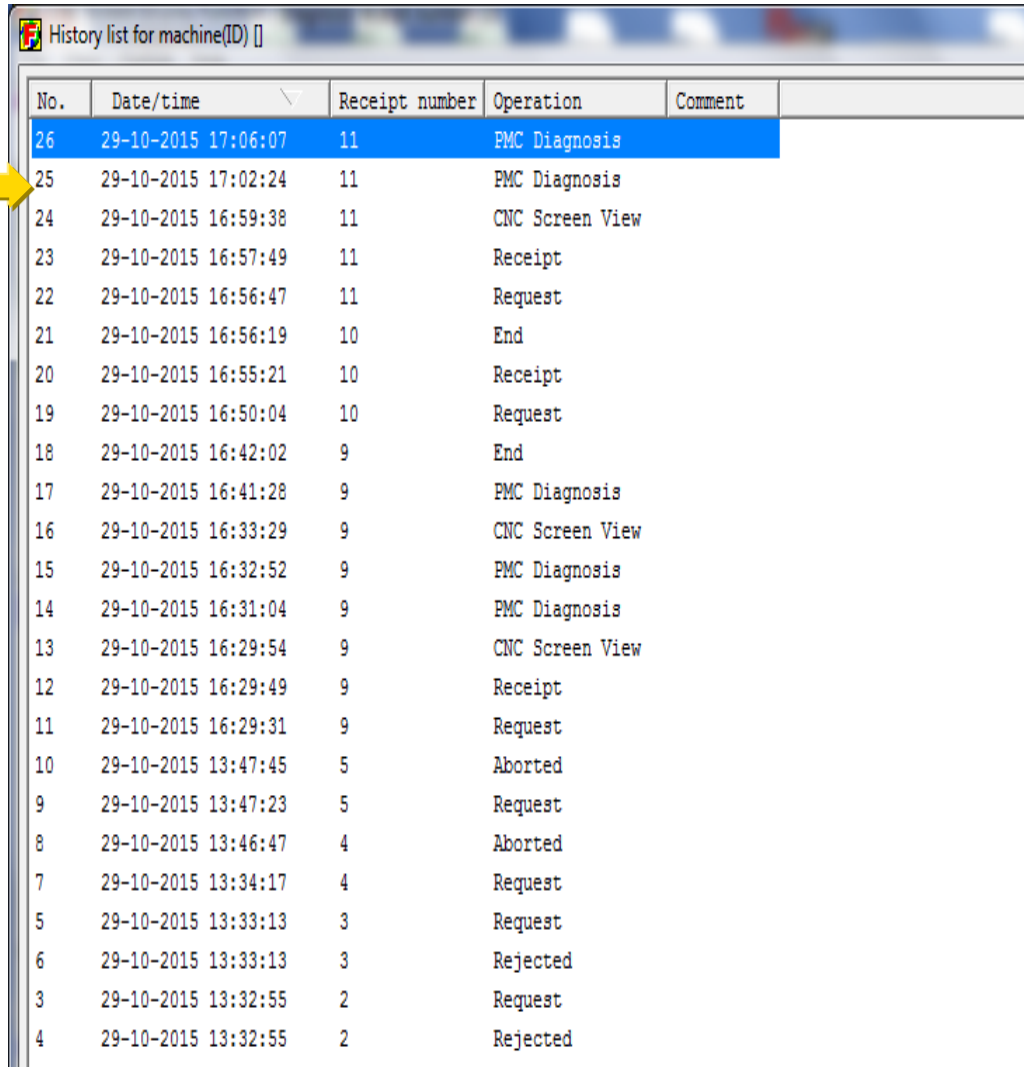


RD Quick NC screen function can be monitored and executed from remote location through MRD

❖ **At most care to be taken during PMC data change from remote location**

MRD – Diagnostic History





No.	Date/time	Receipt number	Operation	Comment
26	29-10-2015 17:06:07	11	PMC Diagnosis	
25	29-10-2015 17:02:24	11	PMC Diagnosis	
24	29-10-2015 16:59:38	11	CNC Screen View	
23	29-10-2015 16:57:49	11	Receipt	
22	29-10-2015 16:56:47	11	Request	
21	29-10-2015 16:56:19	10	End	
20	29-10-2015 16:55:21	10	Receipt	
19	29-10-2015 16:50:04	10	Request	
18	29-10-2015 16:42:02	9	End	
17	29-10-2015 16:41:28	9	PMC Diagnosis	
16	29-10-2015 16:33:29	9	CNC Screen View	
15	29-10-2015 16:32:52	9	PMC Diagnosis	
14	29-10-2015 16:31:04	9	PMC Diagnosis	
13	29-10-2015 16:29:54	9	CNC Screen View	
12	29-10-2015 16:29:49	9	Receipt	
11	29-10-2015 16:29:31	9	Request	
10	29-10-2015 13:47:45	5	Aborted	
9	29-10-2015 13:47:23	5	Request	
8	29-10-2015 13:46:47	4	Aborted	
7	29-10-2015 13:34:17	4	Request	
5	29-10-2015 13:33:13	3	Request	
6	29-10-2015 13:33:13	3	Rejected	
3	29-10-2015 13:32:55	2	Request	
4	29-10-2015 13:32:55	2	Rejected	

Through Diagnostic History Menu:
By clicking on **Diagnostic History** will display various screen on MRD related function executed during trouble shooting process from PC with date and time stamping

Thank you